

SUMIT PANDE

Full Stack Engineer • Backend Specialist • Distributed Systems
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Portfolio | LinkedIn | GitHub

SUMMARY

Full Stack Engineer with **4 years** specializing in backend systems, distributed architecture, and internal platform engineering. Known for owning greenfield products end-to-end — from schema design to production deployment — and for building shared infrastructure that accelerates team delivery. Strong in system design, event-driven architecture, and translating complex requirements into maintainable, scalable systems. Currently targeting Senior SDE/SDE-2 roles in backend-heavy product engineering teams.

EXPERIENCE

Dassault Systèmes

July 2022 - Present

Software Engineering Specialist

Pune, India

- **Led end-to-end development of Skill Portal** — a greenfield enterprise platform now used by 1000+ employees, comprising four integrated modules: ProG, RMG Dashboard, My Competency, and Team Competency.
- **Engineered ProG**, a PDF profile generation system adopted org-wide, **cutting profile creation time by 80%** and saving estimated 200+ hours/month.
- **Built RMG Dashboard** for resource managers to access employee profiles on demand, **improving request turnaround by 50%**.
- **Shipped LLM-powered query engine** integrating Ollama with Node.js for intent detection and dynamic function dispatch — enabling natural language querying over structured data without custom model training, reducing dependency on manual search workflows.
- **Architected and published internal NPM utility library** adopted across 5+ projects, **reducing cross-team code duplication by 40%** — now a core dependency across all project modules.
- Owned end-to-end technical delivery of **Compare-Tool** module led task breakdown, architecture decisions, and code reviews across a 3-engineer team, shipping on committed timelines.

PROJECTS

Nexus | Link

React.js, Node.js, TypeScript, PostgreSQL, Redis, Socket.io, Kafka, Docker

- Architected as distributed system across **6 decoupled services** (API Gateway, Auth, User, Chat, Socket Gateway, Notification) with database-per-service isolation and inter-service communication over kafka — enabling independent deployment and horizontal scaling per service.
- Designed for fault tolerance with **Kafka dead-letter queues** for failed notification delivery, Redis-backed presence tracking with TTL expiry, and cursor-based pagination to handle high-throughput message reads without offset degradation.
- Built a horizontally scalable WebSocket messaging system supporting direct and group conversations using **Redis Pub/Sub** adapter for distributed real-time event fan-out.
- Extended with a **Notification Microservice** consuming Kafka events and routing to email and in-app channels based on user preferences, with delivery tracking and dead-letter retry logic.
- Secured auth with **JWT access tokens** and HTTP-only refresh tokens with rotation and reuse-detection via Redis designed to support stateless horizontal scaling across multiple Auth service instances.
- Modeled a normalized relational schema with composite indexes and cursor-based pagination, optimizing **PostgreSQL** query performance for high-throughput message reads.

SKILLS

Backend: Node.js, Java, Python, TypeScript, REST APIs, Microservices, Distributed Systems, System Design, JWT/Auth Systems, WebSockets

Frontend: JavaScript, HTML, CSS, React.js, Next.js, Vue.js, Zustand

Databases: PostgreSQL, MongoDB, Redis (Pub/Sub, caching), GraphQL, Prisma, Database Design

Messaging: Kafka, Event-Driven Architecture, Pub/Sub, CI/CD

Tools: Docker, Git, GitHub, NPM (package publishing), AWS (EC2, S3), LLM Integration (Ollama)

EDUCATION

Shri Guru Gobind Singhji Institute of Engineering and Technology

Nanded, Maharashtra

CERTIFICATIONS

- CS50W: CS50's Web Programming with Python and JavaScript — **edX**